

Workshop: SNP prioritisation using FINEMAP

07/05/2026

Aim

The aim of this practical is to introduce you to a basic routine for Bayesian finemapping of GWAS summary statistics, using the [FINEMAP](#) tool. Some knowledge of genotype analysis is required to understand the nature of the data being processed, and a basic familiarity with UNIX and R programming is assumed. Other than that, most of the necessary commands to carry out the analyses are specified in detail in the relevant sections.

By the end of the session you will have:

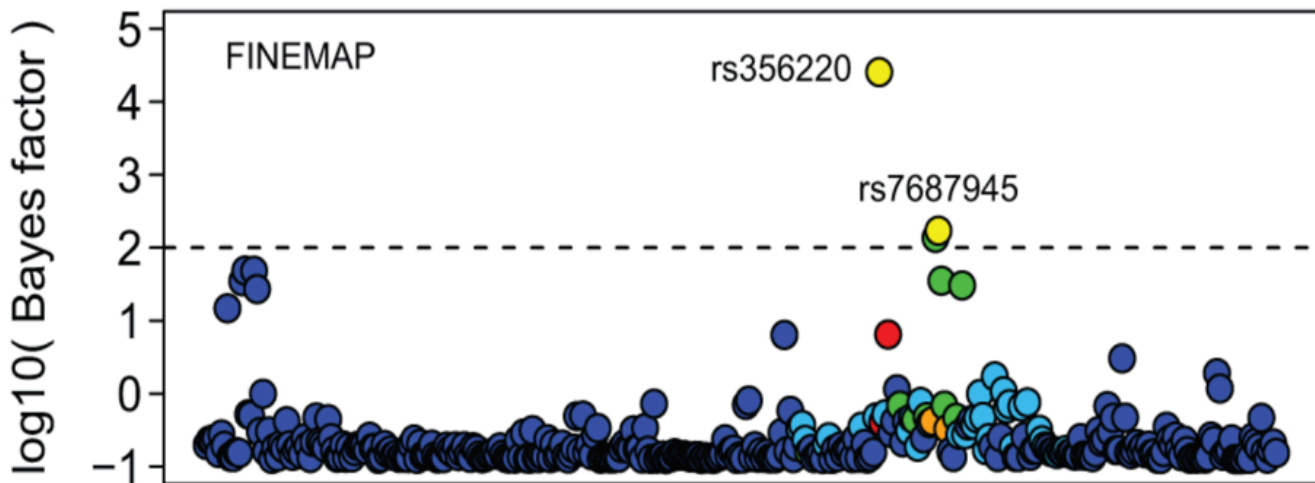
- Prepared GWAS summary statistics for Bayesian finemapping in R, including allele harmonisation against a reference panel, MAF conversion, and effective-sample-size estimation.
- Generated locus-level LD matrices in PLINK and assembled the input files (`.z` , `.ld` , `.infile`) that FINEMAP requires.
- Run FINEMAP under varying assumptions about the maximum number of causal variants (`k`), and interpreted the resulting posterior inclusion probabilities, Bayes factors, and 95% credible sets.
- Contrasted the behaviour of FINEMAP when using a reference-panel LD matrix versus the original study's in-sample LD.
- Visualised finemapping output in LocusZoom, and evaluated the additional information finemapping provides about a GWAS signal.

If you are carrying out these analyses on the Falcon cluster on the day of the lecture, you can enter an interactive session on a dedicated compute node using:

```
salloc --account=SCWF00151_r_anney_149 --cpus-per-task=8 --time=2:00:00  
srun --ntasks=1 --cpus-per-task=8 --pty bash
```

1. Finemapping a single locus with reference LD information

The first exercise has been inspired by the original FINEMAP publication ([Benner et al., 2016](#)), which resolved the *SNCA* locus implicated in Parkinson's disease to two independent signals:



As this result was based on a relatively small study ([UK Parkinson's Disease Consortium and Wellcome Trust Case Control Consortium 2, 2010](#)), it would be nice to test whether it holds with a much larger dataset more recently published ([Nalls et al., 2019](#)). To save time and memory, the summary statistics have been pre-processed to include only chromosome 4.

R operations

The first step will be to load them into the R environment.

```
library(tidyverse)

pdgwas <- readRDS("NallsEtAl2019_excluding23andMe_chr4.rds")
```

If you inspect the `pdgwas` object, you will see that the SNPs have no names and that there are no separate columns for chromosomal/basepair location, both of which FINEMAP requires. The first issue will give us a little bit of work; the second is fortunately trivial. We can also quickly restrict the data to the proximity of the *SNCA* gene (± 500 kb), which fully covers the GWAS locus.

```
pdgwas <- pdgwas |>
  separate(SNP, into = c("CHR", "BP"), sep = ":") |>
  mutate(
    CHR = str_remove(CHR, "chr"),
    CHR = as.integer(CHR),
    BP = as.integer(BP)
  ) |>
  filter(CHR == 4, BP >= 90145250, BP <= 91259466)
```

There are many ways to name a file with unnamed SNPs, but the simplest one is to use a PLINK `.bim` file as a reference. As FINEMAP requires linkage-disequilibrium (LD) information for each SNP, it makes sense to retrieve the names from whatever file we are going to use to estimate these marker-to-marker correlations.

```
chr4snp <- read_table(
  "chr4_1KGp3_eur.bim",
  col_names = c("CHR", "SNP", "cm", "BP", "A1", "A2"),
  col_types = cols_only(CHR = "i", SNP = "c", BP = "i", A1 = "c", A2 = "c")
)
```

Note that, in order to retrieve the SNP names correctly, we need complete information: chromosome, basepair and the two alleles. Because we cannot assume the way alleles are coded as A1/A2 is consistent across files (summary statistics sometimes use the true reference allele as A1, while PLINK assigns A1 to the minor allele unless you're working with dosage data), we need to do the merging in two rounds. Sorting out the names leads to the loss of many SNPs with incompatible alleles (~25% of the data); in a real analysis, this would be a hint that a better procedure is needed (e.g. Genotype Harmonizer; [Deelen et al., 2014](#)), but for this workshop it will do.

```
pdgwas1 <- inner_join(pdgwas, chr4snp, by = join_by(CHR, BP, A1, A2))

pdgwas2 <- inner_join(
  pdgwas, chr4snp,
  by = join_by(CHR, BP, A1 == A2, A2 == A1)
)

pdgwas_rs <- bind_rows(pdgwas1, pdgwas2) |>
  arrange(CHR, BP)

rm(pdgwas1, pdgwas2)
```

The next step is to retrieve the GWAS sample size, which FINEMAP needs. As this meta-analysis has an unequal case/control ratio, it makes sense to compute the so-called *effective sample size* (N_{eff} ; [Willer et al., 2010](#)). This is the sample size that an equally split case/control study would need in order to attain the same statistical power as our GWAS.

However, we need a single value, and here every SNP has been assessed in a different number of samples. To circumvent this, we will use the largest sample size across markers.

```
pdgwas_rs <- pdgwas_rs |>
  mutate(Neff = round(4 / (1 / N_cases + 1 / N_controls)))

max(pdgwas_rs$Neff)
```

Note: In real analyses, you might want to remove markers with low `Neff` before fine-mapping, or use the median `Neff` across all the data as a conservative approximation. If large differences in sample sizes across SNPs are a concern, there are newer algorithms for fine-mapping that can account for per-marker sample sizes (e.g. [Kartau & Pirinen, 2025](#)).

With this data preparation done, we are ready to create the first FINEMAP input file: the `.z` file. This requires us to convert the allele frequencies into MAFs, and to rename a few columns.

```
pdgwas_z <- pdgwas_rs |>
  select(SNP, CHR, BP, A1, A2, freq, b, se) |>
  mutate(freq = if_else(freq > 0.5, 1 - freq, freq)) |>
  rename(
    rsid      = SNP,
    chromosome = CHR,
    position  = BP,
    allele1   = A1,
    allele2   = A2,
    maf       = freq,
    beta      = b
  )

write.table(
  pdgwas_z, "4q22.z",
  col.names = TRUE, row.names = FALSE, quote = FALSE
)
```

Having finished this, we won't need R anymore for this part of the workshop.

Command-line operations

The next step is to calculate the LD from the reference data, using the appropriate PLINK commands. A requirement of FINEMAP is for the LD to be based on exactly the same reference alleles that are specified as A1 in the `.z` file, so notice that we are explicitly forcing this with `--a1-allele`.

```

plink \
  --bfile chr4_1KGp3_eur \
  --a1-allele 4q22.z 4 1 \
  --extract 4q22.z \
  --r square spaces \
  --out 4q22

```

The last file needed is the `.infile`, which can be useful when finemapping several regions at once, although it is not particularly exciting in this practical. This is a plain text document that can be created using simple `echo/print` commands as below; if you are proficient with `vim` or any other command-line editor, please give them a go! We need to create three of these files for our tests, one per value of `k` (the maximum number of causal SNPs). A `for` loop keeps things tidy:

```

for k in 1 2 5; do
  printf 'z;ld;snp;config;cred;log;n_samples\n' > "4q22_k${k}.infile"
  printf
  '4q22.z;4q22.ld;4q22_k%s.snp;4q22_k%s.config;4q22_k%s.cred;4q22_k%s.log;1253
  00\n' \
  "$k" "$k" "$k" "$k" >> "4q22_k${k}.infile"
done

```

The final step is to run the `finemap` program, taking advantage of most default parameters. While these are adequate for nearly all analyses we might undertake during this module, here we also want to test the effect of varying the maximum number of causal SNPs the model will consider (5 by default). Because we have created several `.infile` files, all our output files will have separate names and nothing will be overwritten.

```

for k in 1 2 5; do
  finemap_v1.3.1_x86_64 --log --sss --in-files "4q22_k${k}.infile" \
  --n-causal-snp "$k"
done

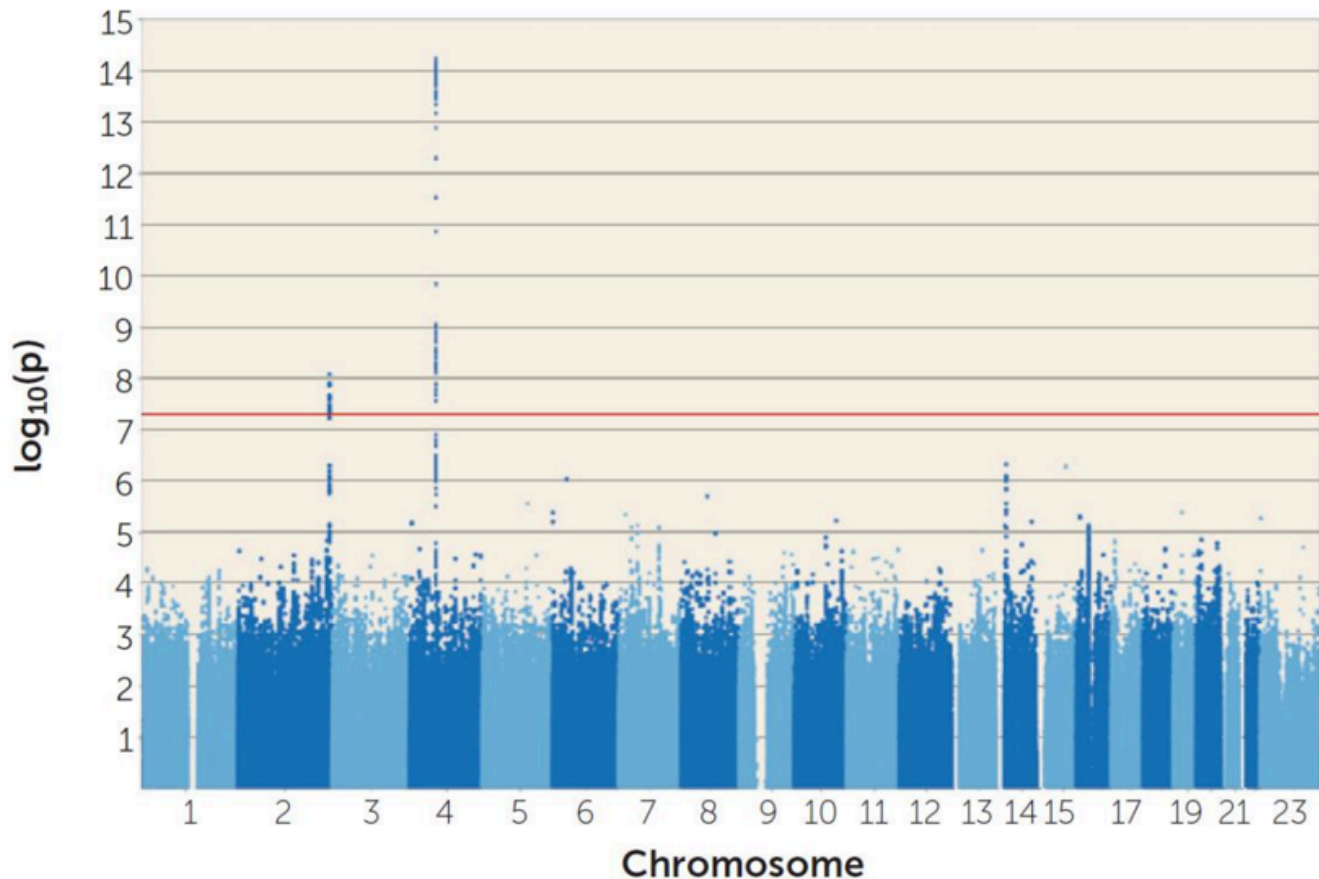
```

You should have results in a few minutes. Now, take a moment to look at the files that have been generated and think about the following questions:

- Check the `.log` files or your command-line output. Based on them, what seems to be the optimal number of causal SNPs (`k`) for this locus?
- Open each `.snp` file with your text editor of choice and look at the posterior probabilities in the `prob` column. What happens as the maximum number of causals increases?
- Look at the contents of the `.cred[k]` files. How many SNPs are in each 95% credible set?

2. Finemapping a single locus with the original LD information

So, that first experience was not particularly productive, was it? Hopefully you have learned a lot and are ready for the next exercise. This example is taken from a publication examining the genetic basis of an antipsychotic (clozapine) plasma concentrations ([Pardiñas et al., 2019](#)). Specifically, the task is to finemap a locus on chromosome 2 where multiple interesting (i.e. protein-coding) variants were found, encompassing several potentially relevant genes.



The full summary statistics are, in this case, pretty straightforward to process. To create the `.z` file, start an R session, load `tidyverse`, and run the following code.

```

library(tidyverse)

norclozglas <- read_table(
  "PardinasEtAl2019_norclozapine_levels.sumstats.mlma.txt.gz"
)

# Original locus: 2:234611523-234676118; here we use ±500 kb around it
norclozglas <- norclozglas |>
  filter(Chr == 2, bp >= 234111523, bp <= 235176118)

norclozglas_z <- norclozglas |>
  select(SNP, Chr, bp, A1, A2, Freq, b, se) |>
  mutate(Freq = if_else(Freq > 0.5, 1 - Freq, Freq)) |>
  rename(
    rsid      = SNP,
    chromosome = Chr,
    position  = bp,
    allele1   = A1,
    allele2   = A2,
    maf       = Freq,
    beta      = b
  )

write.table(
  norclozglas_z, "norcloz_chr2.z",
  col.names = TRUE, row.names = FALSE, quote = FALSE
)

```

Now, quit R and run the following command to calculate the LD matrix and start FINEMAP. Note that in this case we have access to exactly the same data that was used to carry out the GWAS, so the LD we estimate will be a perfect reflection of that used in the original analysis.

```

plink \
  --bfile CLOZUK2_chr2.qc_clozlevels \
  --a1-allele norcloz_chr2.z 4 1 \
  --extract norcloz_chr2.z \
  --r square spaces \
  --out norcloz_chr2

```

The final bits of the analysis are down to you. At this stage you should be able to create the `.infile` needed for an analysis with 5 maximum causal SNPs. After this is finished, think about the following questions:

- Check the `.log` file or your command-line output. Based on it, what seems to be the optimal number of causal SNPs (`k`) for this locus?

- Open the `.cred[k]` file for the optimal `k` and retrieve the 95% credible set. Check Supplementary Table 3 in the original publication: can you find all your SNPs there?
- Load the original summary statistics in R and write a text file with two columns, SNP name and association *P*-value. Then, load the `.snp` file and write another text file with two columns, SNP name and a Bayes-factor-derived score (see hint below). Use these to create regional plots using the [LocusZoom](#) website. Does the topology of the region differ much between analyses? Do the finemapping results clarify the association signal?

Hint. The `.snp` file gives the log10 Bayes factor for each marker (`log10bf`). LocusZoom expects a *P*-value column, so a useful trick is to convert the BF into a "pseudo-*P*" by taking its inverse and capping it at 1; that way SNPs with strong support (large BF) get small values, while SNPs with no support ($BF \leq 1$) get a value of 1. There are two mathematically equivalent transformations to do so ($10^{(-\log_{10}bf)} == 1/BF$), but the second is easier to read.

```
norclozwas_k5 <- read_table("norcloz_chr2_k5.snp") |>
  mutate(
    bf      = 10^log10bf,      # natural-scale Bayes factor
    pseudo_p = pmin(1, 1 / bf) # P-value-like score for LocusZoom
  )
```


References

- Benner, C., Spencer, C. C. A., Havulinna, A. S., Salomaa, V., Ripatti, S., & Pirinen, M. (2016). FINEMAP: efficient variable selection using summary data from genome-wide association studies. *Bioinformatics*, 32(10), 1493–1501.
- Deelen, P., Bonder, M. J., van der Velde, K. J., Westra, H.-J., Winder, E., Hendriksen, D., Franke, L., & Swertz, M. A. (2014). Genotype harmonizer: automatic strand alignment and format conversion for genotype data integration. *BMC Research Notes*, 7, 901.
- Kartau, J., & Pirinen, M. (2025). FINEMAP-miss: fine-mapping genome-wide association studies with missing genotype information. *Bioinformatics*, 41(12), btaf616.
- Nalls, M. A., Blauwendraat, C., Vallerga, C. L., Heilbron, K., Bandres-Ciga, S., Chang, D., Tan, M., Kia, D. A., Noyce, A. J., Xue, A., et al. (2019). Identification of novel risk loci, causal insights, and heritable risk for Parkinson's disease: a meta-analysis of genome-wide association studies. *The Lancet Neurology*, 18(12), 1091–1102.
- Pardiñas, A. F., Nalmpanti, M., Pocklington, A. J., Legge, S. E., Medway, C., King, A., Jansen, J., Helthuis, M., Zammit, S., MacCabe, J., et al. (2019). Pharmacogenomic Variants and Drug Interactions Identified Through the Genetic Analysis of Clozapine Metabolism. *American Journal of Psychiatry*, 176(6), 477–486.
- UK Parkinson's Disease Consortium & Wellcome Trust Case Control Consortium 2 (2010). Dissection of the genetics of Parkinson's disease identifies an additional association 5' of SNCA and multiple associated haplotypes at 17q21. *Human Molecular Genetics*, 20(2), 345–353.
- Willer, C. J., Li, Y., & Abecasis, G. R. (2010). METAL: fast and efficient meta-analysis of genomewide association scans. *Bioinformatics*, 26(17), 2190–2191.